Targeted Adversarial Abundance Attacks for Hyperspectral Image Classification

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***Abstract*—Untargeted adversarial abundance attacks in hyperspectral image (HSI) classification have received growing attention due to their budget efficiency and physical plausibility, which arises from adherence to spectral mixing principles. However, it is difficult to design an abundance-domain perturbation for targeted attacks to simultaneously suppress evidence for the source class while amplifying evidence for a specified target. This article proposes a targeted adversarial abundance attack (TA^3) framework that steers reconstructed spectra toward the decision region of the designated class. The proposed method first applies deep learning-based unmixing to obtain endmembers and abundances without reliance on external spectral libraries. A class-guided perturbation strategy is proposed to enforce abundance-vector similarity and feature-space alignment. Then, a gradient conflict resolution mechanism is proposed to balance multiple loss objectives, which stabilizes updates and maintains small, physically consistent perturbations. Experiments on benchmark HSI datasets demonstrate that the proposed targeted adversarial abundance-attack framework achieves targeted success with improved imperceptibility and small perturbation norms.**

***Index Terms*—Abundance guidance, adversarial attack, feature guidance, gradient interference mitigation, targeted attack.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is to assign a corresponding semantic label to each pixel, enabling the automatic identification and classification of land-cover types. Since each hyperspectral pixel simultaneously retains spatial texture information and provides rich spectral data, the applications of HSI classification have shown great potential across various fields, driving advancements in mineral identification [1], military reconnaissance [2], and others.

Due to their ability to perform nonlinear transformations and to adaptively extract discriminative features, DNNs have been widely applied to HSI classification tasks. They typically extract classification-relevant information by jointly modeling spectral and spatial features. Specifically, a dual-branch convolutional strategy [3] effectively extracts both spectral and spatial features. Then, the Gabor filters are integrated with 2D-CNNs [4] to alleviate overfitting while preserving spatial details. In addition, a deformable HSI classification network was introduced to adaptively sample spatial positions based on feature characteristics [5], thereby showing greater robustness to complex structural patterns. To reduce model complexity and mitigate dependence on large datasets, an end-to-end lightweight 3D-CNN architecture was introduced in [6], which deepens the network while reducing the number of parameters, thereby alleviating overfitting in small-sample scenarios. In recent years, Mamba models [7] have been employed to extract spatial information due to their strength in leveraging contextual dependencies and their low computational complexity [8]. They overcome the limitations of traditional DNNs in capturing long-range features, thereby improving classification accuracy. To further enhance feature discriminability, attention mechanisms have been extensively explored. For instance, a central attention network [9] was proposed to suppress background noise by explicitly weighting the contribution of the central pixel. To address the scale variation of objects, a multiarea target attention framework [10] was introduced to adaptively capture dependencies across different spatial regions. Moreover, SegHSI [11] formulated HSI processing as a semantic segmentation task, demonstrating superior efficiency and robustness even with limited labeled data. Recent studies have further improved the cross-domain generalization capability of deep neural networks in remote sensing tasks. In particular, bi-directional [12] domain adaptation alleviates domain shift caused by variations in scenes and sensors by jointly modeling the relationships between source and target domains. From a domain generalization perspective, single-source domain expansion extends the representational capacity of a source domain to unseen target domains [13], improving robustness in cross-scene classification. In addition, SpectralX [14] introduces a parameter-efficient adaptation strategy that enables remote sensing foundation models originally designed for RGB modalities to generalize across spectral domains.

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Although DNNs perform remarkably well in HSI classification, they are highly sensitive to input data and vulnerable to adversarial attacks [15]. DNNs learn input–output mappings that are significantly discontinuous [16], meaning that even a small perturbation applied to the input can lead to dramatic changes in the output. Adversarial attacks on DNN-based classifiers are generally categorized into untargeted and targeted attacks [17]. Untargeted attacks aim to cause the classifier to output any incorrect labels different from the ground truth [18], [19]. In this hyperspectral domain, a generative adversarial network-based method [18] assigns higher training weights to spectra that are close to the decision boundary or those already misclassified. To overcome limitations of high training costs and the requirement for large amounts of data, a one-pixel attack based on differential evolution [20] was applied in [21], where only a single spectral curve is perturbed to mislead the classifier. In addition, a universal object-level adversarial attack [19] enhances local smoothness of both spatial and spectral perturbations, enabling the generation of large-area, natural-looking adversarial examples with strong transferability across multiple images and models. Targeted attacks, by contrast, aim to mislead the input into a specific, predefined class [22]. This type of attack allows precise control over the classifier's output and poses significant potential risks. An early targeted attempt based on DeepFool can only drive samples to the second-most confident class rather than an arbitrary target [23]. To enable flexible targeting, a genetic and multiobjective evolutionary framework was proposed to search target-specific perturbations, leveraging smoothed key-pixel selection and momentum-augmented mutation [24].

HSI classification is increasingly deployed in safety-critical applications, where stringent security requirements demand improving the reliability and security of underlying classifiers [25]. Nevertheless, most existing adversarial attacks are formulated in the pixel domain by injecting band-wise perturbations into spectral signatures [26], which may violate the physical mixing mechanism of HSIs [27] and thus generate physically implausible spectra [28]. In contrast, abundance-domain perturbations directly manipulate abundance coefficients under nonnegativity and sum-to-one constraints, keeping adversarial samples within the endmember simplex and preserving physical interpretability. This physically consistent formulation can better align with the hyperspectral sensing process and is less likely to be rejected by plausibility-driven defenses. Moreover, since the number of endmembers is typically much smaller than the number of spectral bands [29], [30], abundance-domain attacks are inherently more compact, reducing perturbation dimensionality and associated storage [31]. Existing adversarial attacks in the abundance domain focus on the untargeted paradigm [31] and lack reliable targeted approaches, making it difficult to precisely control the classification outcome after the attack. Compared with untargeted attacks, targeted attacks require simultaneously suppressing the evidence of the original class and enhancing the evidence of the target class, which makes them challenging in terms of optimization and transferability [32]. Motivated by these challenges, this study develops a targeted adversarial attack in the abundance domain for HSI classification. The proposed attack framework reveals that deep classifiers remain highly vulnerable to physically interpretable, abundance-guided targeted perturbations, exposing an overlooked security risk in safety-critical remote-sensing systems. The resulting targeted abundance-level adversarial examples provide a rigorous benchmark for robustness evaluation and defense design, thereby strengthening the foundation for improving the security and reliability of HSI classification [33].

To achieve targeted attacks in the abundance domain, this study proposes a targeted adversarial abundance attacks (TA^3) framework for HSI classification, aiming to mislead the classifier into a specific target class by introducing subtle abundance-level perturbations. Specifically, TA^3 first employs a deep unmixing model to perform hyperspectral unmixing, thereby improving the accuracy of abundance estimation. Based on the similarity between endmember compositions and abundance vectors of the same class, TA^3 designs a dedicated perturbation for each class. Furthermore, a class-guided strategy is introduced to enhance the accuracy of the perturbation direction, which accelerates convergence and helps avoid local optima. Under multigradient guidance, TA^3 adopts a gradient conflict resolution strategy with targeted cross-entropy as the primary optimization direction. Experiments on multiple classifiers demonstrate that TA^3 achieves a high success rate in misleading various models. In summary, the main contributions of this study are as follows.

1) This article proposes an abundance-level targeted attack framework, which introduces abundance-domain perturbations that ensure the physical interpretability of the perturbed hyperspectral data while precisely controlling the classifier's targeted misclassification.
2) This article develops a class-guided perturbation strategy in targeted attacks, which guides the perturbation update direction based on both abundance vectors and the hidden-layer features of the classifier, making the perturbations more representative of the target class, thereby accelerating convergence and improving the attack success rate (ASR).
3) This article introduces a gradient conflict resolution strategy for TA^3, ensuring each update follows a direction that effectively influences the classifier while preserving auxiliary loss effects, thereby reducing optimization oscillations and accelerating convergence.

The remainder of this article is organized as follows. Section II reviews related studies on HSI-targeted adversarial attacks and hyperspectral unmixing. Section III presents and analyzes the proposed TA^3 method in detail. Section IV describes the datasets, experimental settings, and results. Section V concludes the article and provides additional discussions.

II. Background

This section briefly reviews adversarial attacks on HSI and the unmixing process based on deep learning.

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A. Adversarial Attack Methods for HSI

Szegedy et al. [16] showed that adding a human-imperceptible perturbation to a clean input can change a DNN's prediction, introducing the concept of adversarial examples. Under an ℓ_∞-norm constraint, the iterative fast gradient sign method [34] performs iterative projected gradient ascent on the cross-entropy loss with respect to the input. Each step increases the loss for the ground-truth label and then projects the result back to the feasible set, which reduces the classifier's confidence in the true class and promotes misclassification. The update at iteration t is as follows:

$$\boldsymbol{y}^{t+1} = \Pi_{B_\infty(\boldsymbol{y},\epsilon)}\left(\boldsymbol{y}^t + \alpha\,\mathrm{sign}\left(\nabla_{\boldsymbol{y}}\,\mathcal{L}_{\mathrm{CE}}\left(\boldsymbol{y}^t, \mathcal{C}_i\right)\right)\right) \tag{1}$$

where $\boldsymbol{y}$ denotes the original input sample, $\mathcal{C}_i$ represents the ground-truth class label, α denotes the step size of each update, and $\Pi_{B_\infty(\boldsymbol{y},\epsilon)}$ indicates the projection operator that constrains the perturbed result within an ℓ_∞ ball of radius ϵ centered at the clean input $\boldsymbol{y}$. The cross-entropy loss quantifies the discrepancy between the predicted distribution and the true label is described as follows:

$$p_j\,(y) = \frac{e^{z_j(y)}}{\sum_{k=1}^{r} e^{z_k(y)}}, \quad j = 1, \ldots, r, \tag{2}$$

$$\mathcal{L}_{\mathrm{CE}}\,(y, C_i) = -\log p_i\,(y) \tag{3}$$

where r denotes the total number of classes and $z_j(\boldsymbol{y})$ represents the logit score of sample $\boldsymbol{y}$ with respect to class j. Maximizing the cross-entropy with respect to the true label pushes $\boldsymbol{y}$ away from the decision region associated with its ground-truth class by lowering its posterior probability. In the case of targeted attacks, the objective is to misclassify the input into a specific target class $\mathcal{C}_t$. Accordingly, the classifier should increase its confidence for the target class while decreasing its confidence for the true class. Minimizing the cross-entropy loss with respect to the target class thus provides an appropriate initial directional guidance, which can be described as follows:

$$\boldsymbol{y}^{t+1} = \Pi_{B_\infty(\boldsymbol{y},\epsilon)}\left(\boldsymbol{y}^t - \alpha\,\mathrm{sign}\left(\nabla_{\boldsymbol{y}}\,\mathcal{L}_{\mathrm{CE}}\left(\boldsymbol{y}^t, \mathcal{C}_t\right)\right)\right). \tag{4}$$

This update performs signed-gradient descent to minimize the loss while the projection enforces the ℓ_∞ constraint.

B. Hyperspectral Unmixing

The spectral vector of each pixel is considered to be a linear combination of the spectra of multiple materials, accompanied by a certain level of noise in spectral mixing models. Under the linear mixing model, each pixel $\boldsymbol{y}_i$ of HSI $\boldsymbol{Y}$ is expressed as follows:

$$\begin{aligned} &\boldsymbol{y}_i = \boldsymbol{E}\cdot\boldsymbol{a}_i + \boldsymbol{n}_i \\ &\text{s.t.} \sum_{j=1}^{P} a_{i,j} = 1, \quad a_{i,j} \geq 0, \;\; j = 1, 2, \ldots, p \end{aligned} \tag{5}$$

where $\boldsymbol{Y} \in \mathbb{R}^{L*H*W}$ denotes the hyperspectral pixel $\boldsymbol{y}_i \in \mathbb{R}^{L*1}$ and L is the number of spectral bands. The endmember matrix is represented by $\boldsymbol{E} \in \mathbb{R}^{L\times P}$, where P denotes the total number of endmembers. The abundance vector $\boldsymbol{a}_i \in \mathbb{R}^{P*1}$ encodes the proportion of each endmember within the image. The term $\boldsymbol{n}_i \in \mathbb{R}^{P*1}$ corresponds to the noise component.

Hyperspectral unmixing aims to express each pixel spectrum as a linear combination of endmember spectra and to estimate their abundances. We follow the same network configuration as in [35]. The deep unmixing model in the referenced article adopts an autoencoder structure, consisting of a four-layer fully connected encoder network and a single-layer linear decoder network. In the encoder structure, the four fully connected layers progressively reduce the input L-dimensional spectral data, ultimately yielding a P-dimensional abundance vector in the fourth layer. The detailed process is as follows:

$$\begin{aligned} \boldsymbol{v}^{(d)} &= \mathrm{ReLU}\left(\boldsymbol{W}^{(d)\top}\boldsymbol{y} + \boldsymbol{b}^{(d)}\right), \quad d = 1 \\ \boldsymbol{v}^{(d)} &= \mathrm{ReLU}\left(\boldsymbol{W}^{(d)\top}\boldsymbol{v}^{(d-1)} + \boldsymbol{b}^{(d)}\right), \quad d = 2, 3 \\ \boldsymbol{v}^{(d)} &= \boldsymbol{W}^{(d)\top}\boldsymbol{v}^{(d-1)} + \boldsymbol{b}^{(d)}, \quad d = 4 \end{aligned} \tag{6}$$

where $\boldsymbol{W}^{(d)}$ and $\boldsymbol{v}^{(d)}$ denote the weight matrix and output of the dth encoder layer, respectively. The final encoder output $\boldsymbol{v}^{(4)}$ is normalized via Softmax to ensure nonnegativity and a unit-sum constraint, yielding the abundance vector: $\boldsymbol{a}_i = \mathrm{Softmax}(\boldsymbol{v}_i)$. The decoder then reconstructs the spectral pixel from $\boldsymbol{a}_i$ using a single-layer network as follows:

$$\hat{\boldsymbol{y}} = \boldsymbol{W}^\top\hat{\boldsymbol{a}} = \hat{\boldsymbol{E}}\hat{\boldsymbol{a}} \tag{7}$$

where $\boldsymbol{W}$ denotes the weight matrix of the decoder, and the estimated endmembers $\boldsymbol{E}$ are given by its transpose. Subsequently, the encoder and the decoder are jointly optimized. During the optimization of the encoder, the spectral angle between the reconstructed spectrum $\hat{\boldsymbol{y}}_i$ and the ground truth $\boldsymbol{y}_i$ is used as the spectral reconstruction loss $\mathcal{L}_R$. In addition, a regularization term $\mathcal{L}_{\mathrm{SR}}$ is introduced to enforce sparsity. Therefore, the encoder loss is defined as follows:

$$\mathcal{L}_{\mathrm{EN}} = \mathcal{L}_R + \lambda\mathcal{L}_{\mathrm{SR}} \tag{8}$$

$$\begin{aligned} \mathcal{L}_R &= \frac{1}{N}\sum_{i=1}^{N}\mathrm{SAD}\left(\hat{\boldsymbol{y}}_i, \boldsymbol{y}_i\right) \\ &= \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\hat{\boldsymbol{y}}_i^\top\boldsymbol{y}_i}{\left\|\hat{\boldsymbol{y}}_i\right\|_2\left\|\boldsymbol{y}_i\right\|_2}\right) \end{aligned} \tag{9}$$

$$\mathcal{L}_{\mathrm{SR}} = \left\|\hat{\boldsymbol{A}}\right\|_{1/2} = \sum_{i=1}^{N}\sum_{j=1}^{P}\left|\hat{a}_{i,j}\right|^{1/2} \tag{10}$$

where $\boldsymbol{a}_{i,j}$ denotes the abundance of the ith pixel for the jth endmember, $\boldsymbol{N}$ is the number of pixels, and $\boldsymbol{\lambda}$ balances the sparsity term. In the decoder, the spectral reconstruction loss and the minimum volume constraint $\mathcal{L}_{\mathrm{MVC}}$, which encourages a more compact endmember simplex, are used to define the loss as follows:

$$\mathcal{L}_{\mathrm{DE}} = \mathcal{L}_R + \delta\cdot\mathcal{L}_{\mathrm{MVC}} \tag{11}$$

$$\mathcal{L}_{\mathrm{MVC}} = \frac{1}{LP}\sum_{j=1}^{P}\left\|\hat{\boldsymbol{e}}_j - \frac{1}{P}\sum_{i=1}^{P}\hat{\boldsymbol{e}}_i\right\|_2^2 \tag{12}$$

where $\boldsymbol{e}_j \in \mathbb{R}^L$ denotes the learned endmember vector corresponding to the jth endmember and δ is a hyperparameter that scales the minimum-volume regularization. Training proceeds

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via backpropagation with the defined losses, yielding the endmember matrix and the associated abundance vectors.

Adversarial abundance attacks operate in the abundance space and therefore require hyperspectral unmixing to obtain the endmember matrix and abundance vectors. We adopt a deep unmixing model to estimate these representations directly from the input HSI (given the endmember number P), providing reliable abundance estimates without relying on external spectral libraries. This unmixing step forms the foundation of our attack by enabling physically interpretable, abundance-level perturbations and improving attack effectiveness.

III. Methodology

This section first introduces the design principle of class-wise adversarial attacks on HSIs, where distinct perturbations are constructed for each class at the abundance level. Subsequently, a novel formulation termed TA^3 is introduced, and the overall procedure is outlined. Finally, a dedicated class-guided strategy is developed to enhance the directional effectiveness of the perturbations, enabling the classifier to misclassify each pixel into a predefined target class with greater precision.

A. Class-Specific Abundance Perturbation

In conventional pixel-domain attacks, an additive perturbation $\delta_{h,w}$ is applied to each pixel spectrum $\boldsymbol{x}^{(t)} = \boldsymbol{x}^{(t-1)} + \boldsymbol{\delta}_{h,w}$. Although $\boldsymbol{\delta}$ is typically bounded by an ℓ_p norm pixel-wise perturbations are often optimized independently, which can produce fragmented, noise-like patterns across neighboring pixels. Moreover, the update at each step is driven by gradients from individual pixels, leading to low computational efficiency. To address these limitations, the article exploits the similarity of abundance distributions within the same class and learn a class-specific abundance perturbation vector $\boldsymbol{\delta}_i$ for each class. Sharing one perturbation across all pixels of the source class inherently improves spatial consistency, enhances intraclass generalization, and reduces optimization cost. These perturbations collectively form the set $\Delta\boldsymbol{A}$ as follows:

$$\Delta\boldsymbol{A} = \begin{bmatrix} \boldsymbol{\delta}_1^\top \\ \boldsymbol{\delta}_2^\top \\ \vdots \\ \boldsymbol{\delta}_r^\top \end{bmatrix} \in \mathbb{R}^{r\times P}. \tag{13}$$

Compared to pixel-level perturbations, abundance-level ones are lower-dimensional and more physically interpretable. Therefore, the designed perturbation is applied at the abundance level. For an abundance vector $\boldsymbol{a}_{h,w}^{(t)}$ located at (h, w) with ground-truth class i, the abundance-based adversarial vector is added as follows:

$$\boldsymbol{a}_{h,w}^{(t)} = \boldsymbol{a}_{h,w} + \boldsymbol{\delta_i} \tag{14}$$

where $\boldsymbol{a}_{h,w}^{(t)}$ denotes the abundance vector after the tth perturbation. According to (5), the perturbed abundance vector $\boldsymbol{a}_{h,w}^{(t)}$ can be used to reconstruct a new pixel $\boldsymbol{y}_{h,w}^{(t)} = \boldsymbol{E}\cdot\boldsymbol{a}_{h,w}^{(t)} + \boldsymbol{n}_{h,w}$, which in turn forms an adversarial image $\boldsymbol{Y}^{(a)}$. The classifier f_θ then performs inference on this new image $\boldsymbol{Y}^{(a)}$ to produce an updated predicted class $\hat{c}_{h,w}^{(t)}$. If the new predicted class differs from the original one $\hat{c}_{h,w}^{(t)} \neq c_{h,w}$, it indicates that the adversarial patch has successfully fooled the classifier at that pixel.

Algorithm 1 Framework for TA^3

Input: Hyperspectral image $\boldsymbol{Y}$, step size α, max iterations T, classifier f_θ, endmember number P, ground-truth class c, target class $\mathcal{C}_t$
Output: Adversarial image $\boldsymbol{Y}^{(a)}$

1. Perform hyperspectral unmixing on $\boldsymbol{Y}$ using a deep learning model to obtain abundance matrix $\boldsymbol{A}$;
2. Split all abundance vectors corresponding to class c into multiple mini-batches, where each batch is denoted as $\{x_1, x_2, \ldots, x_b\}$;
3. Initialize class-specific perturbations vector for class c as $\boldsymbol{\delta}_c = 0$;
4. Optimize $\boldsymbol{\delta}_c$ through the iterative process in **Algorithm 2**, and obtain the final class-specific perturbation $\boldsymbol{\delta}_c^{(T)}$;
5. $\boldsymbol{\delta}_c = \boldsymbol{\delta}_c^{(T)}$;
6. $\boldsymbol{A}^{(a)} = \boldsymbol{A} + \boldsymbol{\delta}_c$ for (h, w) such that label $(h, w) \in \mathcal{P}_c$;
7. Reconstruct image $\boldsymbol{Y}^{(a)}$ using Eq. (5);
8. **return** $\boldsymbol{Y}^{(a)}$;

B. Framework of TA^3 for HSI Classification

As previously mentioned, image adversarial attacks aim to perturb the abundance matrix $\boldsymbol{A}$ by adding $\Delta\boldsymbol{A}$, so that the HSI reconstructed from the modified abundances can mislead the classifier. Conventional untargeted strategies often yield inconsistent predictions for spatially contiguous pixels belonging to the same class, which appear as discontinuities in the classification map. To address this limitation and enable precise control of the output label, we propose a TA^3 method that forces the classifier to consistently misclassify a given source class c into a specific target class $\mathcal{C}_t$, which is formulated as follows:

$$f_\theta\left(\boldsymbol{a}_{h,w} + \boldsymbol{\delta}_c\right) = \mathcal{C}_t, \quad \text{for } (h, w) \in \mathcal{P}_c \tag{15}$$

where $f_\theta(\cdot)$ denotes the hyperspectral classifier and $\boldsymbol{a}_{h,w}$ represents an abundance vector corresponding to the original class c of location (h, w).

The framework is show in Fig. 1 and the overall workflow of the TA^3 framework for HSIs is illustrated in Algorithm 1. It consists of three main stages: spectral unmixing, abundance perturbation computation, and adversarial reconstruction and evaluation. The stages proceed sequentially and are summarized as follows.

1) Spectral Unmixing: Unlike traditional unmixing methods, the proposed TA^3 framework employs a deep encoder-decoder structure, avoiding the assumptions of pure pixel endmembers and linear mixing, thereby improving abundance estimation in complex data. The encoder is composed of a four-layer network that compresses the input hyperspectral pixel of L spectral bands into an abundance vector of dimension P as (6).

The abundance vector $\boldsymbol{a}$ can be obtained by normalizing the output of the fourth layer of the encoder $\boldsymbol{a}_i = \text{Softmax}(\boldsymbol{v}_i)$. The

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decoder then optimizes the estimation of the abundance vector $\boldsymbol{a}$, and the corresponding endmember matrix can be described as follows:

$$\frac{1}{N}\sum_{(h,w)\in \boldsymbol{Y}} \boldsymbol{E} = \boldsymbol{y}_{(h,w)} \cdot \boldsymbol{a}_{(h,w)}^{-1}. \tag{16}$$

2) Abundance Perturbation Computation: Given a source class c to be attacked and a target class $\mathcal{C}_t$, a corresponding perturbation vector $\boldsymbol{\delta}_c$ is calculated using the method described in Algorithm 2. The perturbation is added to all pixels belonging to class c.

3) Adversarial Reconstruction and Evaluation: Based on (5), the perturbed abundance matrix $\boldsymbol{A}$ is used to reconstruct an adversarial HSI $\boldsymbol{Y}$, which is then fed into the classifier to obtain prediction results and evaluate the ASR, Specifically, ASR measures the proportion of pixels originally belonging to the source class c that are successfully misclassified into the designated target class $\mathcal{C}_t$

$$\text{ASR} = \frac{N_{c\to\mathcal{C}_t}}{N_c} \times 100\% \tag{17}$$

where $N_{c\to c_t}$ denotes the number of pixels from class c that are misclassified as $\mathcal{C}_t$ and N_c represents the total number of pixels in class c.

C. Class-Guided Perturbation Strategies

To mitigate the significant disparity between ground-truth classes and target classes, TA^3 proposes a class-guided optimization method. This approach progressively adjusts the abundance vector toward the target class with each perturbation step, which enhances attack accuracy and reduces the requisite number of perturbation updates.

Based on the abundance vector $\boldsymbol{a}_{h,w}$ and the classifier's predicted confidence $\pi_{h,w,i}$ of each pixel, a targeted abundance vector for class i can be computed. Specifically, the confidence $\pi_{h,w,i}$ serves as a weight $w_{h,w}$ for the corresponding abundance vector $\boldsymbol{a}_{h,w}$. By performing a weighted average across all pixels, a representative abundance vector that characterizes class i is obtained

$$\boldsymbol{a_i} = \frac{\sum_{(h,w)\in\mathcal{P}_i} w_{(h,w)} \cdot \boldsymbol{a}_{(h,w)}}{\sum_{(h,w)\in\mathcal{P}_i} w_{(h,w)}} \tag{18}$$

where $\boldsymbol{a}_i$ denotes the abundance vector for the ith class and $\sum_{(h,w)\in\mathcal{C}_i}$ represents all the pixel points belonging to the ith class. The abundance cosine similarity loss $\mathcal{L}_{\text{ACS}}$ measures the difference between the abundance vector of a pixel under the current perturbation $\boldsymbol{a}_{h,w}^{(t)}$ and the targeted abundance vector $\boldsymbol{a}_t$ of the target class t

$$\begin{aligned}\mathcal{L}_{\text{ACS}} &= \frac{1}{N}\sum_{(h,w)\in\mathcal{P}_i}\left(1-\cos\theta_{h,w}^{(t)}\right)\\ &= \frac{1}{N}\sum_{(h,w)\in\mathcal{P}_i}\left(1-\frac{\boldsymbol{a}_{h,w}^{(t)}\cdot\boldsymbol{a}_t}{\left\|\boldsymbol{a}_{h,w}^{(t)}\right\|_2\cdot\|\boldsymbol{a}_t\|_2}\right)\end{aligned} \tag{19}$$

where $\sum_{(h,w)\in\mathcal{C}_i}$ denotes the set of all pixels originally belonging to class i that are intended to be misclassified as class t and N represents the total number of such pixels. Subsequently,

Algorithm 2 Category-Specific Guidance Strategy for Perturbation Optimization

Input: Original hyperspectral image $\boldsymbol{Y}$, endmember matrix $\boldsymbol{E}$, target class $\mathcal{C}_t$, Class c with abundance set $\mathcal{A}_c = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_N\}$, Pre-trained classifier f_θ, batch size b, learning rate α, total iterations T

Output: Final perturbation $\boldsymbol{\delta}_c^{(T)}$

[Only in first iteration]:
Compute class probability map using f_θ and determine weights $w_{h,w,\mathcal{C}_t}$
Compute targeted abundance vector $\boldsymbol{a}_{\mathcal{C}_t}$ and targeted feature vector $\boldsymbol{z}_{\mathcal{C}_t}$
Initialize perturbation: $\boldsymbol{\delta}_c^{(0)} = 0$
for $t = 1$ **to** T **do**
  Partition $\mathcal{A}_c$ into mini-batches $\{\mathcal{B}_1, \ldots, \mathcal{B}_K\}$ of size b; set $\boldsymbol{\delta}_c^{(t,0)} \leftarrow \boldsymbol{\delta}_c^{(t-1)}$.
  for $k = 1$ **to** K **do**
    Select batch $\mathcal{B}_k = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_b\}$.
    for $i = 1$ **to** b **do**
      Add perturbation: $\boldsymbol{a}_i^{(t)} = \boldsymbol{a}_i + \boldsymbol{\delta}_c^{(t,k-1)}$
      Reconstruct image: $\boldsymbol{y}_i^{(t)} = \boldsymbol{E} \cdot \boldsymbol{a}_i^{(t)}$
      Extract feature $\boldsymbol{z}_i^{(t)}$ from the penultimate layer of f_θ
    Compute all losses: $\mathcal{L}_{\text{TCE}}, \mathcal{L}_{\text{ACS}}, \mathcal{L}_{\text{FCS}}, \mathcal{L}_{\text{ANC}}, \mathcal{L}_{\text{ASR}}$ (see Eq. (19), (22), (24), (26), (27))
    Compute gradients: $\boldsymbol{g}_{\text{TCE}}, \boldsymbol{g}_i$ (see Eq. (20), (23), (25)), where $i \in \mathcal{I} \setminus \{\text{TCE}\}$
    foreach $i \in \mathcal{I} \setminus \{TCE\}$ **do**
      if $\boldsymbol{g}_i \cdot \boldsymbol{g}_{TCE} < 0$ **then**
        $\boldsymbol{g}_i^{\perp} = \boldsymbol{g}_i - \frac{\boldsymbol{g}_i \cdot \boldsymbol{g}_{\text{TCE}}}{\|\boldsymbol{g}_{\text{TCE}}\|^2} \cdot \boldsymbol{g}_{\text{TCE}}$
      else
        $\boldsymbol{g}_i^{\perp} = \boldsymbol{g}_i$
    Aggregate gradients: $\boldsymbol{g}_{\text{target}} = \sum_{i\in\mathcal{I}} w_i \cdot \boldsymbol{g}_i^{\perp}$
    $\boldsymbol{\delta}_c^{(t,k)} \leftarrow \boldsymbol{\delta}_c^{(t,k-1)} + \alpha \cdot \text{sign}(\boldsymbol{g}_{\text{target}})$
  Set $\boldsymbol{\delta}_c^{(t)} \leftarrow \boldsymbol{\delta}_c^{(t,K)}$.
return $\boldsymbol{\delta}_c^{(T)}$

The gradient of the abundance cosine similarity loss $\mathcal{L}_{\text{ACS}}$ is calculated as follows:

$$\begin{aligned}\nabla_{\delta_i}\mathcal{L}_{\text{ACS}} = -\frac{1}{N}\sum_{(h,w)\in\mathcal{P}_i}&\frac{\boldsymbol{a}_t}{\|\boldsymbol{a}_t\|_2\left\|\boldsymbol{a}_{h,w}^{(t)}\right\|_2}\\ &\cdot\left(1-\frac{\boldsymbol{a}_{h,w}^{(t)}}{\left\|\boldsymbol{a}_{h,w}^{(t)}\right\|_2^2}\boldsymbol{a}_{h,w}^{(t)}\right).\end{aligned} \tag{20}$$

The gradient is computed to update the perturbation, thereby encouraging the current abundance vector to align more closely with the target class t.

The feature vector refers to the output of the intermediate hidden layer of the classifier when performing classification

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on the current pixel. In TA^3, the penultimate layer is selected to retain higher-level information while avoiding the nondifferentiability issue associated with the final probabilistic output layer. The targeted feature vector for class i is computed using a weighted averaging approach, where the weights $w_{h,w}$ follow the same formulation as in the targeted abundance vector. The targeted feature vector $\boldsymbol{z}_i$ of the class i and classifier f_θ can be formally defined as follows:

$$\boldsymbol{z}_i^{(f_\theta)} = \frac{\sum_{(h,w)\in\mathcal{P}_i} w_{h,w} \cdot \boldsymbol{z}_{h,w}}{\sum_{(h,w)\in\mathcal{P}_i} w_{h,w}}. \tag{21}$$

Then, the feature cosine similarity loss $\mathcal{L}_{\text{FCS}}$ is formulated to measure the difference between the feature vector under the current perturbation $\boldsymbol{z}_{h,w}^{(t)}$ and the targeted feature vector $\boldsymbol{z}_t$ of the target class t, which uses the cosine spectral angle loss as follows:

$$\begin{aligned}\mathcal{L}_{\text{FCS}} &= \frac{1}{N}\sum_{(h,w)\in\mathcal{P}_i}\left(1-\cos\theta_{h,w}^{(t)}\right)\\ &= \frac{1}{N}\sum_{(h,w)\in\mathcal{P}_i}\left(1-\frac{\boldsymbol{z}_{h,w}^{(t)}\cdot\boldsymbol{z}_t}{\left\|\boldsymbol{z}_{h,w}^{(t)}\right\|_2\cdot\|\boldsymbol{z}_t\|_2}\right).\end{aligned} \tag{22}$$

Subsequently, the gradient of the feature cosine similarity loss $\mathcal{L}_{\text{FCS}}$ is calculated as follows:

$$\begin{aligned}\nabla_{\delta_i}\mathcal{L}_{\text{FCS}} = -\frac{1}{N}\sum_{(h,w)\in\mathcal{P}_i}&\frac{\boldsymbol{z}_t}{\|\boldsymbol{z}_t\|_2\left\|\boldsymbol{z}_{h,w}^{(t)}\right\|_2}\\ &\cdot\left(1-\frac{\boldsymbol{z}_{h,w}^{(t)}}{\left\|\boldsymbol{z}_{h,w}^{(t)}\right\|_2^2}\boldsymbol{z}_{h,w}^{(t)}\right).\end{aligned} \tag{23}$$

By leveraging the abundance and feature vectors, the perturbed pixels are guided to exhibit greater similarity to those of the target class. The target cross-entropy loss further directs the classifier's output by maximizing its confidence in the target class t. Specifically, it minimizes the negative log-likelihood of the predicted probability for the target class t

$$\mathcal{L}_{\text{TCE}} = -\sum_{h=1}^{H}\sum_{w=1}^{W}\log\left(\max\left(\pi_{h,w,t}, e^{-\tau/r}\right)\right) \tag{24}$$

where $\pi_{h,w,t}$ denotes the probability output by the classifier for class t at spatial position (h,w), and a loss threshold τ is introduced to prevent $\log(\pi_{h,w,t})$ from becoming too small or even approaching negative infinity. The corresponding gradient is computed as follows:

$$\nabla_{\delta_i}\mathcal{L}_{\text{TCE}} = \frac{1}{N}\sum_{(h,w)\in P_i}\boldsymbol{E}^\top\cdot\nabla_{\boldsymbol{y}_{h,w}}\mathcal{L}_{\text{TCE}} \tag{25}$$

where $\boldsymbol{E}$ denotes the endmember matrix as defined in (5) and $\boldsymbol{y}_{h,w}$ denotes the hyperspectral pixel at the location (h,w).

D. Gradient Conflict Resolution Strategy

In addition to class-guided updates that drive the prediction toward the target class, constraints are imposed to preserve imperceptibility and to keep abundance values nonnegative. an adversarial nonnegativity constraint $\mathcal{L}_{\text{ANC}}$ is introduced to penalize negative values, and an adversarial sum-to-one constraint $\mathcal{L}_{\text{ASC}}$ is employed to quantify the deviation between the perturbed and original abundance vectors, which is consistent with the formulation used in traditional nontargeted attacks

$$\mathcal{L}_{\text{ANC}} = \frac{1}{H*W*P}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{k=1}^{P}\left(\min\left(a_{h,w,k}^{(t)},0\right)\right)^2 \tag{26}$$

$$\mathcal{L}_{\text{ASC}} = \frac{1}{H*W}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(\sum_{k=1}^{P}a_{h,w,k}^{(t)}-\sum_{k=1}^{P}a_{h,w,k}\right)^2 \tag{27}$$

where P denotes the number of endmembers, $a_{h,w,k}^{(t)}$ denotes the abundance value of the kth endmember at spatial position (h,w) under the current image with perturbation $\boldsymbol{a}_{h,w}^{(t)}$, while $a_{h,w,k}$ denotes the original abundance value.

Joint optimization of multiple objectives often yields opposing gradients, which can cause oscillation and impede convergence. Therefore, a gradient conflict resolution strategy is introduced to explicitly steer optimization toward the target class. The target cross-entropy loss $\mathcal{L}_{\text{TCE}}$, which directly influences the classifier's output, is treated as the primary direction. If the gradient of another loss function conflicts with that of the target cross-entropy loss: $\boldsymbol{g}_{\text{TCE}}\cdot\boldsymbol{g}_i < 0$, its gradient is projected onto the orthogonal complement of the target cross-entropy gradient to mitigate interference

$$\boldsymbol{g}_i^{\perp} = \boldsymbol{g}_i - \frac{\boldsymbol{g}_i\cdot\boldsymbol{g}_{\text{TCE}}}{\left\|\boldsymbol{g}_{\text{TCE}}\right\|^2}\boldsymbol{g}_{\text{TCE}} \tag{28}$$

where $\boldsymbol{g}_i^{\perp}$ denotes the projection of the conflicting gradient $\boldsymbol{g}_i$ onto the orthogonal complement of the target cross-entropy gradient $\boldsymbol{g}_{\text{TCE}}$. Subsequently, individual weights are assigned to each loss term to reflect their relative importance according to specific requirements. The overall loss function is defined as follows:

$$\mathcal{L}_{\text{target}} = \sum_{i\in\mathcal{I}} w_i\cdot\mathcal{L}_i \tag{29}$$

where $\mathcal{I} = \{\text{TCE}, \text{ACS}, \text{FCS}, \text{ANC}, \text{ASC}\}$. Algorithm 2 illustrates the process of generating category-targeted perturbation updates for the batch associated with the target class t in a targeted attack scenario.

IV. Experimental Studies

In this section, we conduct extensive experiments to evaluate the performance and characteristics of the proposed TA^3 attack. We first provide an overview of the experimental setup, then analyze the influence of key experimental parameters on the results and identify the optimal configuration. Finally, we compare our method with several state-of-the-art targeted attack approaches.

A. Experimental Settings

This section introduces the datasets used in the experiments and the victim classification networks targeted by the TA^3 attack. In addition, it presents the training setup for the classifiers and the evaluation metrics used for assessing the attacks.

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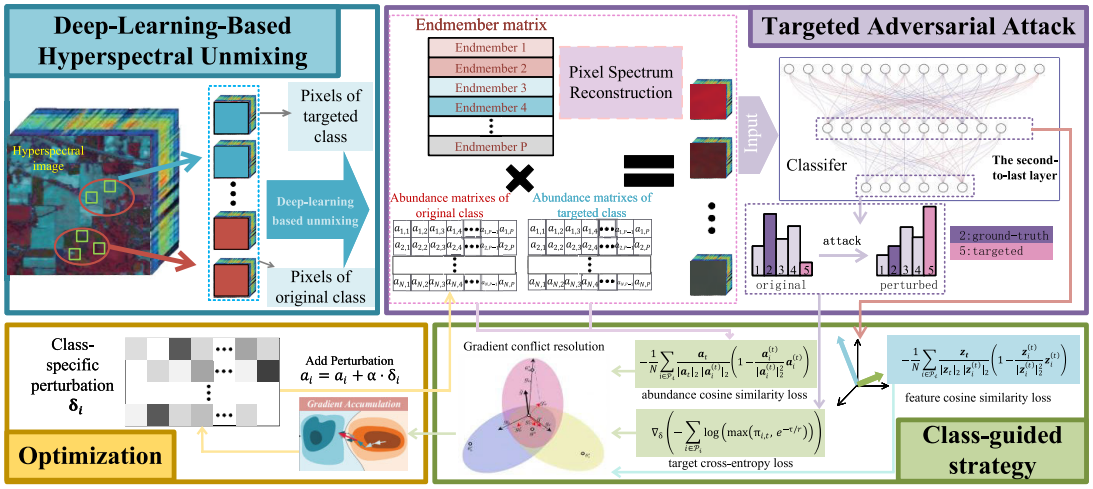


Fig. 1. Architecture of the TA^3 framework for HSI classification. A deep unmixing model first decomposes the input HSI into endmember and abundance matrices. Perturbations are then added to the abundance vectors, and the reconstructed spectra are used to compute the three guidance losses defined in TA^3. Finally, the perturbation vectors are updated via gradient optimization.

1) Datasets: This study is conducted on three publicly available hyperspectral datasets: Indian Pines (IP),[1] Salinas (SA),[2] and Pavia University (PU).[2] The false color image and ground-truth maps of all three datasets are shown in the supplementary materials.

IP Datasets: Collected by AVIRIS over agricultural areas in northwestern Indiana, USA, the IP datasets contain 145 × 145 pixels with 220 spectral bands spanning the wavelength range of 0.4–2.5 μm. After removing water absorption bands, 200 bands are retained. The ground truth includes 16 mainly agricultural classes.

Salinas Datasets: The SA datasets have high spatial resolution and large scene scale, which were acquired by the AVIRIS sensor over the Salinas Valley in California, USA. It has a spatial size of 512 × 217 pixels and includes 224 spectral bands covering the 0.4–2.5-μm range. After discarding water absorption bands, 204 bands are used. The ground truth is annotated into 16 classes of crops and land-cover types.

PU Datasets: The PU datasets are a typical urban remote sensing scene acquired by the ROSIS sensor over Pavia in northern Italy. It consists of 610 × 340 pixels with 115 spectral bands covering the wavelength range of 0.43–0.86 μm. After removing noisy bands, 103 bands are commonly used in experiments. The ground truth contains nine urban land-cover classes, such as asphalt, meadows, trees, metal roofs, and shadows. This dataset is widely used due to its representativeness as a typical urban remote sensing scene.

2) Victim Classification Network: We evaluated the effectiveness of the proposed attack method on eight well-trained classifiers, namely VGG11 [36], VGG16 [36], ResNet18 [37], ResNet50 [37], AlexNet [38], DenseNet121 [39], 3D-DenseNet [40], Lightweight-VGG [41], MambaHSI [42], and GraphGSTHSI [43]. 3D-DenseNet employs 3-D convolutions to meet the processing requirements of volumetric data. Its architecture leverages dense connections to achieve cross-layer feature aggregation, enabling low-level detail features to be fully integrated with high-level semantic features. Lightweight-VGG performs spectral dimensionality reduction through band selection and principal component analysis and employs an improved lightweight VGG backbone, which effectively balances classification accuracy and computational efficiency. MambaHSI employs selective state space layers to model hyperspectral cubes as spectral sequences with embedded spatial context, enabling the network to capture long-range spectral dependencies while maintaining linear computational complexity with respect to spectral length. GraphGSTHSI constructs a graph-structured spectral–spatial representation, where pixels or superpixels are treated as nodes and their relationships as edges, and leverages graph-based aggregation to exploit nonlocal contextual information and reduce spectral redundancy, thereby enhancing classification robustness in complex scenes.

3) Attacking Method: To assess its effectiveness in HSI classification, the TA^3 method is compared with nine state-of-the-art adversarial attack baselines, including CGNC [44], AnyAttack [45], Logit [46], SASD-WS [47], SS-FGSM [19], MI-SS-FGSM [19], AdaEA [48], SMER [49], and T-I-FGSM [50]. CGNC [44] leverages CLIP by conditioning a perturbation generator on the text embedding of the target class, producing fine-grained, class-directed perturbations. AnyAttack [45] targets vision–language models by pretraining a noise generator with self-supervised embedding alignment between noise-augmented and clean images, followed by task-specific fine-tuning. Logit [46] uses a target–nontarget logit margin instead of cross-entropy to mitigate gradient saturation. SASD-WS [47] improves surrogate transferability via sharpness-aware self-distillation and then performs targeted generation by weight-space scaling. SS-FGSM and MI-SS-FGSM [19] extend FGSM/MI-FGSM updates [17], [51] with

[1]https://doi.org/10.4231/R7RX991C

[2]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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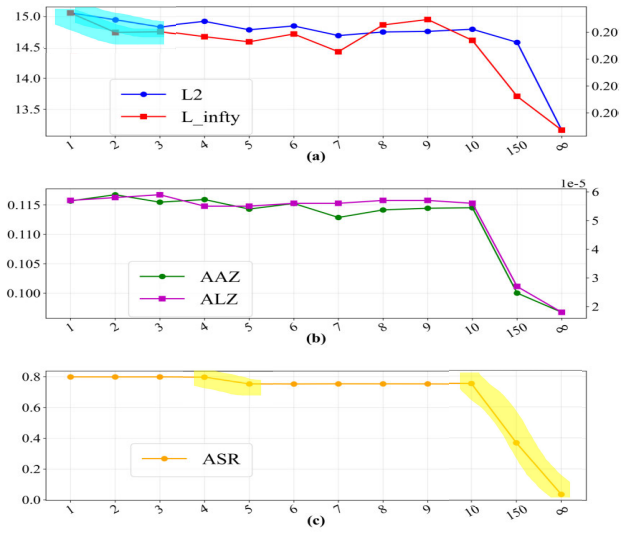


Fig. 2. Metric trends under different cross-entropy loss thresholds (top to bottom) perturbation imperceptibility metrics (L_2, L_∞), physical consistency metrics (AAZ and ALZ), and ASR. Yellow regions mark significant ASR drops; blue regions mark noticeable decreases in other metrics without ASR changes. a) Imperceptibility metrics. (b) Physical consistency metrics. (c) ASR.

spectral–spatial and object-level consistency; we adapt them to targeted attacks by optimizing target-class cross-entropy. AdaEA [48] and SMER [49] are ensemble-based attacks that stabilize updates through effectiveness-aware gradient reweighting with discrepancy filtering and diversity-preserving reinforcement reweighting, respectively. T-I-FGSM [50] iteratively minimizes the target-class loss along the input gradient, pushing samples toward the specified target region.

4) Training Settings of the Unmixing Model: In the unmixing model, the input hyperspectral data are normalized to the range [0, 1], and the unmixing autoencoder is trained in a pixel-wise manner with a patch size of 1×1. The model is trained end-to-end for 80 epochs with a batch size of 128. We use the Adam optimizer with a tiered learning-rate setting, where the encoder learning rate is 3×10^{-4} and the decoder learning rate is 1×10^{-4}. For training stability, we adopt a StepLR scheduler with a decay factor of 0.8 every 20 epochs.

5) Evaluating Metrics: We evaluated the performance of the TA^3 method under all perturbation update strategies from three perspectives: attack effectiveness, perturbation imperceptibility, and physical consistency. Specifically, attack effectiveness was measured by the ASR. Perturbation imperceptibility was quantified by the L_2 norm, which reflects the overall perturbation magnitude, and the L_∞ norm, which represents the maximum perturbation amplitude. Physical consistency was assessed based on the nonnegativity and sum-to-one properties of abundance values, using abundance-level zeroing (ALZ) and abundance average zeroing (AAZ), respectively.

The L_2 and L_∞ norm can be expressed as follows:

$$L_2 = \frac{1}{N}\sum_{i=1}^{N} \|\Delta \boldsymbol{a}_i\|_2 \tag{30}$$

$$L_\infty = \frac{1}{N}\sum_{i=1}^{N} \|\Delta \boldsymbol{a}_i\|_\infty = \frac{1}{N}\sum_{i=1}^{N} \max_j \left|\Delta a_i^j\right| \tag{31}$$

where N denotes the total number of test samples, $\Delta \boldsymbol{a}_i$ denotes the adversarial perturbation applied to the abundance vector of the ith sample, and Δa_i^j denotes the perturbation added to the jth element of the abundance vector of the ith sample.

The ALZ and AAZ can be expressed as follows:

$$\mathrm{ALZ} = \frac{1}{N \cdot P}\sum_{i=1}^{N}\sum_{p=1}^{P} \max\left(0, \ -\left(a_{i,p} + \Delta a_{i,p}\right)\right) \tag{32}$$

$$\mathrm{AAZ} = \frac{1}{N}\sum_{i=1}^{N}\left|\sum_{p=1}^{P}\left(a_{i,p} + \Delta a_{i,p}\right) - 1\right| \tag{33}$$

where P denotes the number of endmembers, $a_{i,p}$ denotes the original abundance value of the pth endmember in the sample, and $\Delta a_{i,p}$ represents the adversarial perturbation added to this abundance value.

B. Experiments on the Proposed TA^3

In this section, we investigate the effect of the parameter τ, the variations of the loss weights w_{ACS} and w_{FCS}, and batch size b on the attack performance.

1) Test of the Parameter τ: This section examines the effect of thresholding the targeted cross-entropy loss on attack performance, using the IP, SA, and PU datasets. The threshold τ is assigned values over $[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 150, \infty]/1/r$, where r denotes the number of classes. Specifically, $r = 16$ for the IP and SA datasets, and $r = 9$ for the PU datasets. By comparing the variations of evaluation metrics under different thresholds, the results are presented in Fig. 2. At excessively large thresholds (up to ∞), L_2, L_∞, AAZ and ALZ decrease but ASR drops sharply. Under a low threshold, the loss decreases only after the target-class probability becomes high and optimization is steered to further raise it. In contrast, a higher threshold causes the classifier to pay more attention to minimizing perturbation magnitude and preserving physical properties of abundances, even when the confidence for the target class remains relatively low. According to Fig 2, when the threshold is set between 1 and 4, the ASR remains nearly unchanged while both L_2, L_∞ decrease. The ASR decreases compared to the former range when the threshold is set between 5 and 10, whereas L_2, L_∞, AAZ, and ALZ show no significant differences. Therefore, the optimal threshold value selected in this study is $\tau = 4$.

2) Test of the Batch Size b: Since the TA^3 method updates the class-level abundance perturbation vector using a mini-batch strategy, all abundance vectors belonging to the same class c are divided into several batches of size b. For each batch, the loss and gradient are computed, and the perturbation is updated sequentially. To evaluate the impact of batch size on attack performance, we conduct a systematic comparison with $b \in [2, 4, 8, 16, 32, 64, 128, 256, 512]$, while keeping all other settings unchanged. The experiments show that increasing b from 2 to 64 yields a marked ASR gain. The larger batch provides a more accurate gradient direction through sample averaging and reduces outlier impact. Beyond $b = 64$, the

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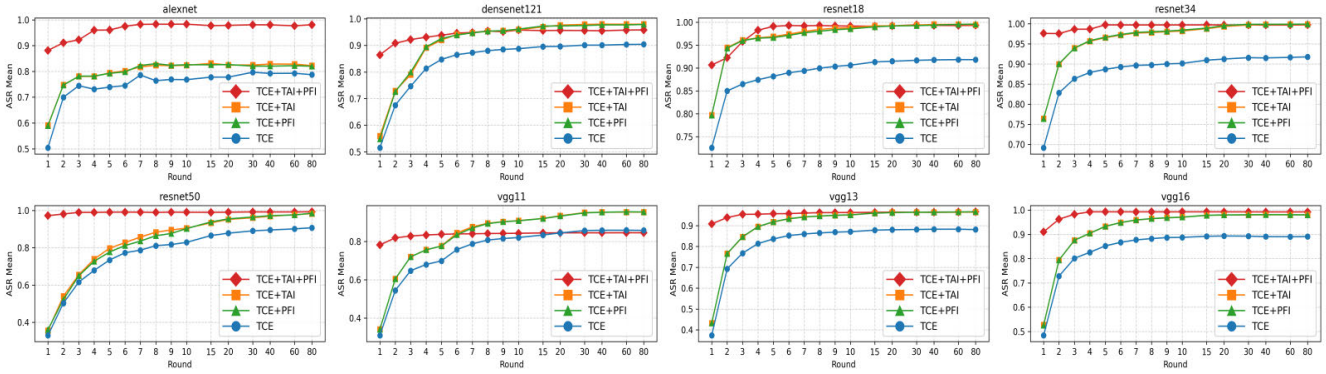


Fig. 3. ASR variations over attack epochs for eight models on the PU datasets. The subplots are arranged from left to right, top to bottom, in the order of AlexNet, DenseNet, ResNet-18, ResNet-50, VGG-11, VGG-16, 3D-DenseNet, and lightweight-VGG. Each curve corresponds to a different loss configuration (TCE, TCE + ACS, TCE + FCS, and TCE + ACS + FCS).

ASR improvement becomes marginal and gradually plateaus. In addition, there is almost no noticeable difference in the physical consistency and imperceptibility metrics. Given the computational cost associated with larger batch sizes, b is set to 64.

3) Ablation Studies: To verify the effect of each loss term in TA^3, we design a series of ablation experiments on different combinations of loss functions and evaluate the performance in terms of ASR, L_2, L_∞, AAZ, and ALZ across multiple classification models. Experimental results indicate that variants using ACS or FCS alone without TCE underperform. TCE is therefore included in all configurations, and the effects of ACS and FCS are compared. The attack accuracy under each configuration is assessed and presented in Fig. 3 and Tables I and II. It should be noted that each loss term is associated with a weighting coefficient specified as (TCE:ACS:FCS:ANC:ASC = 4:2.5:2.5:1:1) (e.g., $\mathcal{L}_{\text{TCE}}$ actually denotes $w_{\text{TCE}} \cdot \mathcal{L}_{\text{TCE}}$), and $\mathcal{L}_{\text{ASC}}$ and $\mathcal{L}_{\text{ANC}}$ are additionally included in all of the above configurations to ensure the physical consistency and imperceptibility of perturbations. Fig. 3 illustrates the evolution of the ASR over iterations. It can be observed that when TCE, ACS, and FCS act jointly, the overall ASR increases and the convergence speed becomes faster across multiple models. TCE directly constrains the classifier's output toward the target label, whereas ACS and FCS adjust the gradient in the abundance and feature spaces. When the three terms are applied jointly, both AAZ and L_∞ decrease relative to the baselines, indicating that the generated perturbations are smaller in magnitude and less destructive to the original physical semantics and thus making the adversarial patches more imperceptible. It is worth noting that applying either ACS or FCS alone can improve the final ASR to some extent, but the generated adversarial patches do not exhibit a significant change in their impact on the original image. This pattern indicates that both ACS and FCS act to refine the composite search direction, yielding more stable, target-aligned updates.

C. Attack Performance

In this section, we evaluate the performance of the proposed TA^3 attack method on eight different neural network

TABLE I

EVALUATION ON THE IP DATASETS USING A DIFFERENT LOSS CONFIGURATION

Method	Metric	AX	D121	R18	R50	V11	V16	3DD	LwV	Mamba	Graph
all	L_2	14.61	10.54	7.870	9.458	15.32	13.84	7.902	14.25	10.03	8.007
	$L_\infty(\times 10^{-2})$	19.62	18.78	17.31	16.86	20.93	20.90	17.43	20.63	17.14	17.28
	$ALZ(\times 10^{-6})$	75.00	30.00	19.00	32.00	38.00	64.00	15.00	66.00	18.00	12.00
	$AAZ(\times 10^{-2})$	10.88	7.284	5.285	6.260	11.06	11.17	5.469	11.09	6.973	5.552
	$ASR(\%)$	90.46	98.05	99.61	98.32	95.26	98.61	99.80	95.31	88.39	90.18
A	L_2	20.95	12.42	12.02	10.42	16.73	15.68	10.04	17.15	10.11	8.976
	$L_\infty(\times 10^{-2})$	22.01	20.77	20.85	17.95	22.00	21.64	19.36	22.10	18.14	17.99
	$ALZ(\times 10^{-6})$	66.00	52.00	42.00	31.00	59.00	58.00	41.00	72.00	17.00	14.00
	$AAZ(\times 10^{-2})$	16.83	8.259	7.609	6.885	12.71	12.44	6.136	13.54	7.102	6.188
	$ASR(\%)$	88.74	96.99	99.33	98.62	95.20	96.56	99.52	95.36	84.68	87.60
F	L_2	12.17	7.680	5.011	6.587	10.44	9.518	5.150	9.699	10.06	7.871
	$L_\infty(\times 10^{-2})$	18.86	16.08	10.99	14.23	18.42	18.05	11.70	17.29	16.65	17.13
	$ALZ(\times 10^{-6})$	64.00	18.00	1.000	10.00	42.00	43.00	2.000	35.00	19.00	10.00
	$AAZ(\times 10^{-2})$	8.106	4.168	0.7421	4.142	6.402	6.353	1.223	5.459	6.978	5.325
	$ASR(\%)$	86.36	97.41	99.29	97.98	94.80	96.06	99.15	95.24	84.12	88.69
cls	L_2	22.08	12.56	11.80	10.64	16.57	15.28	9.890	16.55	10.03	8.376
	$L_\infty(\times 10^{-2})$	22.16	20.64	20.59	17.47	21.91	21.13	19.73	22.11	17.65	17.21
	$ALZ(\times 10^{-6})$	105.0	70.00	60.00	23.00	97.00	58.00	57.00	84.00	18.00	9.000
	$AAZ(\times 10^{-2})$	18.26	8.367	7.418	7.133	12.53	11.77	6.161	13.05	6.975	5.358
	$ASR(\%)$	83.47	97.95	98.51	96.86	92.48	94.08	97.81	91.44	80.64	82.18

TABLE II

EVALUATION ON THE SA DATASETS USING A DIFFERENT LOSS CONFIGURATION

Method	Metric	AX	D121	R18	R50	V11	V16	3DD	LwV	Mamba	Graph
all	L_2	5.868	4.918	3.018	3.222	4.310	4.564	3.105	5.110	6.478	6.300
	$L_\infty(\times 10^{-2})$	7.044	7.144	5.513	6.108	6.160	6.540	5.847	6.706	7.446	7.307
	$ALZ(\times 10^{-6})$	1.000	2.000	2.000	0.000	0.000	3.000	1.000	0.000	0.000	0.000
	$AAZ(\times 10^{-2})$	5.028	3.925	1.392	1.671	2.772	3.059	1.306	3.810	5.957	5.429
	$ASR(\%)$	88.93	89.98	99.83	99.65	83.25	98.19	99.76	91.65	75.85	78.95
A	L_2	8.951	6.348	6.647	5.835	6.795	6.902	5.878	7.146	6.425	6.258
	$L_\infty(\times 10^{-2})$	7.817	7.804	7.810	7.784	7.807	7.814	7.772	7.815	7.278	7.223
	$ALZ(\times 10^{-6})$	1.000	2.000	3.000	0.000	0.000	3.000	2.000	0.000	0.000	0.000
	$AAZ(\times 10^{-2})$	8.514	5.249	5.595	4.725	5.717	6.135	4.638	6.322	5.872	5.350
	$ASR(\%)$	84.39	88.66	99.74	99.42	80.48	97.17	99.83	89.72	68.11	64.53
F	L_2	8.400	6.129	6.361	5.444	6.275	6.015	5.382	6.813	6.555	6.471
	$L_\infty(\times 10^{-2})$	7.809	7.776	7.774	7.738	7.767	7.761	7.793	7.805	7.478	7.266
	$ALZ(\times 10^{-6})$	1.000	2.000	2.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
	$AAZ(\times 10^{-2})$	7.726	4.966	5.180	4.273	5.276	5.344	3.955	5.876	6.027	5.617
	$ASR(\%)$	84.98	88.68	99.65	99.01	80.68	96.38	99.67	89.67	64.80	70.65
cls	L_2	6.883	5.631	4.780	4.563	5.544	5.698	4.613	5.981	6.317	6.041
	$L_\infty(\times 10^{-2})$	7.724	7.477	7.342	7.283	7.666	7.672	7.386	7.645	7.146	7.059
	$ALZ(\times 10^{-6})$	1.000	2.000	1.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
	$AAZ(\times 10^{-2})$	6.166	4.603	3.572	3.361	4.514	4.817	3.346	5.223	5.807	5.045
	$ASR(\%)$	80.51	88.63	98.49	98.69	80.17	93.59	97.34	86.71	55.52	59.66

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TABLE III

COMPARISON OF TA^3 AGAINST COMPETING ATTACKS ACROSS TEN CLASSIFIERS ON THE PU DATASETS. ABBREVIATIONS: AX: ALEXNET, D121: DENSENET-121, R18: RESNET-18, R50: RESNET-50, V11: VGG-11, V16: VGG-16, 3DD: 3D-DENSENET, LWV: LIGHTWEIGHT-VGG, MAM: MAMBAHSI, AND GRA: GRAPHGSTHSI

Attack	Metric	AX	D121	R18	R50	V11	V16	3DD	LWV	Mam	Gra
TA^3	L_2	6.883	5.631	4.780	4.563	5.544	5.698	4.613	5.981	6.478	6.300
	$L_\infty(\times10^{-2})$	7.720	7.480	7.340	7.280	7.670	7.670	7.390	7.640	7.446	7.307
	$ALZ(\times10^{-6})$	1.000	2.000	1.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
	$AAZ(\times10^{-2})$	6.166	4.603	3.572	3.361	4.514	4.817	3.346	5.223	5.957	5.429
	$ASR(\%)$	**85.51**	**88.63**	**99.49**	**98.69**	**80.17**	**97.59**	**99.34**	**88.71**	**75.85**	**78.95**
CGNC [44]	L_2	9.244	8.676	8.229	8.543	9.663	9.042	8.741	9.033	7.640	6.964
	$L_\infty(\times10^{-2})$	13.19	13.17	13.18	13.05	13.13	13.17	13.16	13.12	12.86	12.91
	$ALZ(\times10^{-6})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	10.56	9.963	9.563	9.508	10.23	10.66	9.674	10.44	7.682	7.754
	$ASR(\%)$	15.35	7.070	39.63	8.770	22.45	9.200	29.39	14.04	0.040	1.012
AnyAttack [45]	L_2	13.38	11.85	11.33	10.93	12.84	12.38	10.88	13.06	10.03	8.468
	$L_\infty(\times10^{-2})$	13.18	13.20	12.75	13.22	13.23	13.09	13.12	13.22	12.28	12.39
	$ALZ(\times10^{-6})$	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	1.000	1.000
	$AAZ(\times10^{-2})$	15.98	13.77	13.45	13.47	14.70	15.96	13.36	15.51	9.554	9.386
	$ASR(\%)$	80.99	85.92	88.97	86.93	72.69	88.97	87.83	75.92	50.72	67.32
Logit [46]	L_2	15.39	12.21	13.75	13.04	13.55	12.58	12.40	13.98	10.62	8.836
	$L_\infty(\times10^{-2})$	13.23	13.23	13.23	13.23	13.23	13.23	13.23	13.23	12.26	12.95
	$ALZ(\times10^{-6})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	18.29	14.57	16.99	15.79	16.58	16.36	15.90	17.44	11.24	9.507
	$ASR(\%)$	77.93	82.65	88.47	81.54	58.30	74.83	87.80	72.86	44.95	60.00
SASD-WS [47]	L_2	18.05	14.67	13.04	17.21	11.11	13.87	16.81	15.32	7.874	10.36
	$L_\infty(\times10^{-2})$	12.43	11.52	11.68	12.06	11.43	11.98	10.96	11.42	10.48	11.28
	$ALZ(\times10^{-6})$	1.000	2.000	2.000	2.000	1.000	1.000	2.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	23.88	20.04	20.97	21.44	20.12	22.76	19.93	19.64	16.02	19.64
	$ASR(\%)$	0.080	8.010	10.29	10.02	0.000	0.000	3.110	0.000	0.000	0.000
MI-SS-FGSM [19]	L_2	9.776	8.768	6.104	7.863	10.67	6.878	5.825	10.39	10.00	8.220
	$L_\infty(\times10^{-2})$	12.16	12.40	11.98	12.84	12.67	7.040	12.05	12.54	11.96	12.00
	$ALZ(\times10^{-6})$	1.000	1.000	1.000	1.000	1.000	0.0000	1.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	10.30	8.407	6.016	8.997	9.189	5.697	5.328	9.162	8.841	6.490
	$ASR(\%)$	82.05	84.00	88.38	85.46	61.26	82.18	86.17	73.90	50.73	74.16
SS-FGSM [19]	L_2	12.63	10.40	11.23	11.54	11.08	11.53	11.89	12.64	9.937	9.236
	$L_\infty(\times10^{-2})$	13.23	13.23	12.95	13.23	13.23	13.23	13.23	13.23	13.00	13.05
	$ALZ(\times10^{-6})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	15.11	13.78	12.32	13.85	15.04	15.69	15.55	16.14	10.41	10.22
	$ASR(\%)$	52.60	30.98	20.85	11.17	25.82	26.57	32.04	19.64	32.26	20.29
AdaEA [48]	L_2	6.575	7.607	5.565	6.706	6.849	7.177	5.523	6.288	7.369	6.011
	$L_\infty(\times10^{-2})$	11.86	12.40	11.78	12.24	12.53	12.38	11.44	11.71	12.21	11.75
	$ALZ(\times10^{-6})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	7.381	8.969	5.757	8.763	7.665	9.219	5.920	7.153	8.837	5.110
	$ASR(\%)$	25.30	29.42	31.13	32.04	31.94	30.14	30.14	24.81	2.097	2.742
SMER [49]	L_2	6.581	7.543	5.671	6.636	6.839	7.180	5.551	6.303	7.383	5.991
	$L_\infty(\times10^{-2})$	11.80	12.39	11.66	12.32	12.43	12.29	11.56	11.71	12.31	11.89
	$ALZ(\times10^{-6})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	7.414	8.891	5.926	8.678	7.717	9.268	5.956	7.226	8.835	5.128
	$ASR(\%)$	80.98	86.03	86.41	84.74	78.22	80.44	86.95	78.44	69.36	74.77
T-I-FGSM [50]	L_2	9.927	8.954	8.739	9.439	7.853	8.043	8.593	8.821	5.303	6.814
	$L_\infty(\times10^{-2})$	7.820	7.820	7.820	7.820	7.820	7.780	7.820	7.820	9.827	9.706
	$ALZ(\times10^{-6})$	2.000	1.000	2.000	2.000	0.000	1.000	0.000	1.000	1.000	1.000
	$AAZ(\times10^{-2})$	9.310	7.973	8.252	8.995	6.924	7.681	7.780	7.779	7.353	5.060
	$ASR(\%)$	81.26	42.79	77.65	45.28	44.06	76.24	45.55	56.76	11.87	7.500

TABLE IV

COMPARISON OF TA^3 AGAINST COMPETING ATTACKS ACROSS TEN CLASSIFIERS ON THE IP DATASETS. ABBREVIATIONS: AX: ALEXNET, D121: DENSENET-121, R18: RESNET-18, R50: RESNET-50, V11: VGG-11, V16: VGG-16, 3DD: 3D-DENSENET, LWV: LIGHTWEIGHT-VGG, MAM: MAMBAHSI, AND GRA: GRAPHGSTHSI

Attack	Metric	AX	D121	R18	R50	V11	V16	3DD	LWV	Mam	Gra
TA^3	L_2	20.95	12.42	12.02	10.42	16.73	15.68	10.04	17.15	10.03	8.007
	$L_\infty(\times10^{-2})$	22.01	20.77	20.85	21.64	22.00	21.64	19.36	22.10	17.14	17.28
	$ALZ(\times10^{-6})$	6.600	5.200	4.200	3.100	5.900	5.800	4.100	7.200	18.00	12.00
	$AAZ(\times10^{-2})$	16.83	8.259	7.609	6.885	12.71	12.44	6.136	13.54	6.973	5.552
	$ASR(\%)$	**88.74**	**96.99**	99.33	**98.62**	**95.20**	**96.56**	**99.52**	**95.99**	**88.39**	**90.18**
CGNC [44]	L_2	18.77	14.16	15.19	13.58	16.66	16.80	14.71	16.72	14.00	12.78
	$L_\infty(\times10^{-2})$	21.97	20.95	21.59	20.38	21.86	21.94	21.41	21.93	20.48	20.40
	$ALZ(\times10^{-6})$	8.800	10.70	9.300	9.400	10.30	8.500	10.00	8.700	61.00	88.00
	$AAZ(\times10^{-2})$	15.25	9.987	11.59	10.68	13.16	13.53	11.68	13.33	10.31	9.108
	$ASR(\%)$	5.269	1.200	0.488	7.630	2.742	1.200	2.732	2.111	3.483	12.59
AnyAttack [45]	L_2	17.49	12.35	9.582	8.533	18.27	18.96	9.257	18.63	11.49	9.517
	$L_\infty(\times10^{-2})$	19.05	17.72	14.83	13.77	20.40	20.41	15.24	20.41	17.15	16.43
	$ALZ(\times10^{-6})$	5.000	2.500	4.000	4.000	6.300	4.800	4.000	4.000	42.00	28.00
	$AAZ(\times10^{-2})$	12.11	8.495	6.931	5.411	13.32	14.62	6.820	14.28	7.959	7.035
	$ASR(\%)$	85.07	89.46	85.33	88.85	88.77	80.67	86.30	89.74	81.00	83.99
Logit [46]	L_2	30.29	17.57	20.38	19.16	24.90	25.10	22.51	23.92	8.390	8.200
	$L_\infty(\times10^{-2})$	22.19	21.74	21.94	20.93	22.19	22.19	21.80	22.19	15.14	17.58
	$ALZ(\times10^{-6})$	14.20	6.900	9.300	7.500	7.100	7.600	11.00	6.900	14.00	14.00
	$AAZ(\times10^{-2})$	27.36	12.19	15.03	13.46	20.51	22.27	16.08	20.06	6.227	5.442
	$ASR(\%)$	68.34	78.74	**99.97**	45.17	74.79	89.12	86.58	87.49	61.96	85.85
SASD-WS [47]	L_2	34.79	27.46	33.34	34.76	28.25	33.21	27.20	32.22	19.02	24.40
	$L_\infty(\times10^{-2})$	22.14	20.78	21.09	20.43	19.89	20.54	19.86	19.96	14.12	19.69
	$ALZ(\times10^{-6})$	27.90	28.10	17.70	25.30	1.900	10.40	4.600	23.90	96.00	171.0
	$AAZ(\times10^{-2})$	34.94	31.59	32.06	32.05	28.75	30.97	28.79	29.52	13.85	27.65
	$ASR(\%)$	0.000	0.527	0.000	0.000	0.000	0.000	0.390	0.000	0.000	0.000
MI-SS-FGSM [19]	L_2	14.66	9.116	7.343	14.22	9.804	8.907	8.445	9.167	11.33	9.375
	$L_\infty(\times10^{-2})$	18.73	16.34	13.93	19.87	16.50	16.09	14.26	15.47	17.25	16.95
	$ALZ(\times10^{-6})$	8.700	1.200	1.000	5.500	2.600	3.500	1.000	2.800	30.00	28.00
	$AAZ(\times10^{-2})$	8.882	4.883	3.335	9.405	4.948	4.589	4.193	4.582	7.407	4.934
	$ASR(\%)$	78.56	88.66	89.54	67.17	85.41	86.68	88.96	86.79	77.97	82.73
SS-FGSM [19]	L_2	25.31	20.76	21.19	19.14	19.71	23.25	21.32	20.29	18.53	17.64
	$L_\infty(\times10^{-2})$	22.19	22.19	22.19	22.19	22.19	22.19	22.19	22.19	21.89	21.71
	$ALZ(\times10^{-6})$	7.400	8.900	9.400	8.500	9.200	7.700	8.600	7.800	44.00	47.00
	$AAZ(\times10^{-2})$	19.91	16.74	17.39	15.89	16.37	19.22	17.36	17.06	16.09	15.29
	$ASR(\%)$	4.537	11.40	17.49	31.42	20.06	45.89	15.26	23.60	6.635	15.88
AdaEA [48]	L_2	10.82	9.750	6.803	11.96	10.52	10.29	6.627	10.64	11.95	12.46
	$L_\infty(\times10^{-2})$	19.48	19.13	14.51	21.00	18.19	18.97	14.23	18.75	19.52	19.29
	$ALZ(\times10^{-6})$	1.140	0.930	0.130	0.970	0.960	0.760	0.030	0.860	105.0	112.0
	$AAZ(\times10^{-2})$	7.835	6.326	2.365	9.013	6.793	8.175	2.443	7.350	7.813	8.004
	$ASR(\%)$	40.98	27.31	27.31	49.18	51.91	40.98	35.52	31.42	0.332	0.673
SMER [49]	L_2	11.09	9.968	6.649	11.53	10.51	10.32	6.619	10.67	11.65	12.38
	$L_\infty(\times10^{-2})$	19.53	19.05	14.19	20.92	18.23	19.00	14.33	18.71	19.39	19.27
	$ALZ(\times10^{-6})$	1.140	0.870	0.100	0.960	0.990	0.770	0.030	0.890	96.00	112.0
	$AAZ(\times10^{-2})$	8.052	6.454	2.348	8.445	6.841	8.117	2.447	7.337	7.578	7.819
	$ASR(\%)$	75.00	78.17	88.05	86.81	74.83	78.01	89.59	78.02	73.35	89.91
T-I-FGSM [50]	L_2	26.77	22.26	22.65	22.24	21.79	21.86	23.84	21.74	10.92	9.283
	$L_\infty(\times10^{-2})$	22.16	22.19	22.19	22.19	22.19	22.07	22.19	22.19	15.12	16.38
	$ALZ(\times10^{-6})$	1.170	1.200	0.6400	1.350	0.870	0.570	0.850	0.730	23.00	126.0
	$AAZ(\times10^{-2})$	27.59	18.11	19.64	18.59	18.23	21.64	19.32	19.75	6.718	9.707
	$ASR(\%)$	53.52	50.45	59.83	60.23	75.74	88.49	58.63	86.63	3.327	15.57

classifiers, built in Section IV-B. Each original class is randomly assigned to a different target class to demonstrate its effectiveness across various class mappings, and two such mapping sets are constructed for the attacks. In addition, we compare the TA^3 method against nine state-of-the-art adversarial attacks originally designed for HSI classification or image classification, including CGNC [44], AnyAttack [45], Logit [46], SASD-WS [47], SS-FGSM [19], MI-SS-FGSM [19], AdaEA [48], SMER [49], and T-I-FGSM [50]. For a fair comparison with our targeted setting, untargeted methods (SS-FGSM and MI-SS-FGSM) are adapted to their targeted counterparts following their original frameworks. All competing targeted methods are implemented in the abundance space for hyperspectral data. Results on the IP and PU datasets are summarized in Tables III and IV, where each entry is averaged over all source–target mappings.

As shown in Tables III and IV, our targeted attack method TA^3 achieves a notably higher ASR than others across all eight classifiers, demonstrating its clear advantage in attack capability. In addition, owing to our designed abundance and feature guidance strategies, the perturbed abundance vectors are driven closer to the real abundance of the target class, which helps better preserve the nonnegativity and sum-to-one properties of abundances. Consequently, the AAZ and ALZ values exhibit a certain degree of reduction. Since TA^3 introduces two additional loss terms and a gradient conflict resolution strategy, the perturbation-strength constraint enforced by L_{ASC} is partially diluted, leading to a modest increase in perturbation magnitude. Nevertheless, this increase remains within an acceptable range and does not affect the method's ability to reveal the vulnerability of deep learning classifiers to targeted attacks and their high sensitivity to input data.

To further provide an intuitive visualization and validation of the attack results, we assign a unique color to each class and compare the spatial distribution of classifier outputs before and after the attack. Figs. 4–6 present the performance of the eight models on the IP, SA, and PU datasets, respectively, where the combination of TCE + ACS + FCS is adopted. In these figures, the color at each spatial location indicates the predicted class label of the corresponding pixel. The first subfigure shows the

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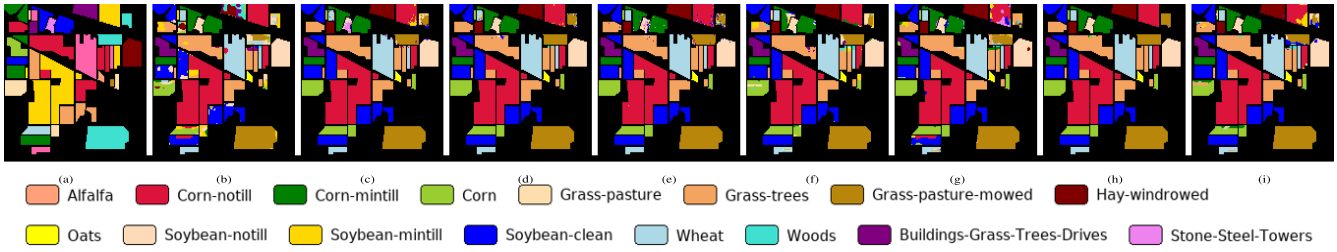


Fig. 4. Ground-truth class map of the IP scene and the corresponding classification maps produced by different networks under the proposed TA^3 targeted abundance attack, where most pixels are successfully driven into the designated target class. (a) Original. (b) AlexNet. (c) DenseNet-121. (d) ResNet-18. (e) ResNet-50. (f) VGG-11. (g) VGG-16. (h) 3D-DenseNet. (i) Lightweight-VGG.

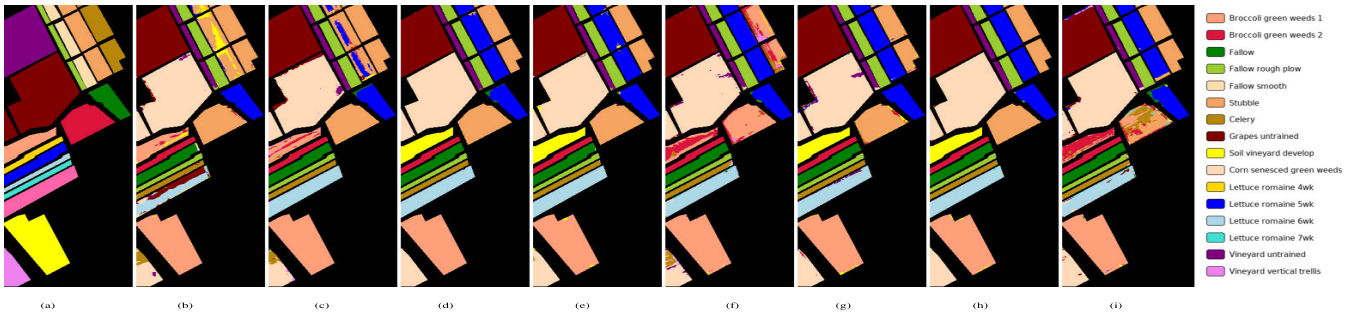


Fig. 5. Ground-truth class map of the SA scene and the corresponding classification maps produced by different networks under the proposed TA^3 targeted abundance attack, where most pixels are successfully driven into the designated target class. (a) Original. (b) AlexNet. (c) DenseNet-121. (d) ResNet-18. (e) ResNet-50. (f) VGG-11. (g) VGG-16. (h) 3D-DenseNet. (i) Lightweight-VGG.

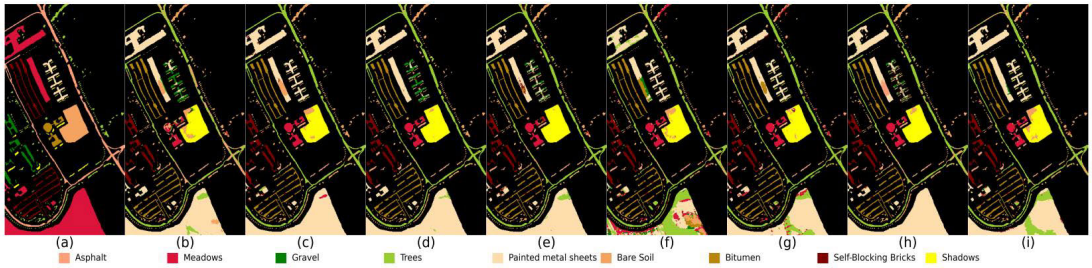


Fig. 6. Ground-truth class map of the PU scene and the corresponding classification maps produced by different networks under the proposed TA^3 targeted abundance attack, where most pixels are successfully driven into the designated target class. (a) Original. (b) AlexNet. (c) DenseNet-121. (d) ResNet-18. (e) ResNet-50. (f) VGG-11. (g) VGG-16. (h) 3D-DenseNet. (i) Lightweight-VGG.

ground-truth label map, while the remaining subfigures depict the per-pixel classification results of different classifiers under attack. It can be observed that, after the attack, most regions are rendered in the color of the target class, which implies that the majority of pixels have been successfully misclassified into the target class by the classifiers.

V. Concluding Remarks

This article proposes a novel TA^3 framework for HSI classification, termed TA^3. The proposed approach not only overcomes the limitations of existing pixel-level attacks that often distort the physical properties of hyperspectral data but also fills the gap of targeted attacks at the abundance level. By incorporating deep learning-based hyperspectral unmixing, the framework reduces the difficulty of obtaining abundance vectors and endmember matrices while improving estimation accuracy. Furthermore, class-specific perturbations are designed to effectively balance computational complexity and attack effectiveness. Lastly, TA^3 employs a carefully crafted loss function to guide perturbation updates precisely and effectively, which accelerates convergence and enhances classification accuracy. Experimental results on three benchmark datasets demonstrate that TA^3 achieves higher accuracy and lower perturbation magnitude than several state-of-the-art targeted attacks when applied to HSI classification. Future research will focus on the transferability of adversarial examples generated by TA^3 and their applicability in black-box attack scenarios.

References

[1] N. Agrawal and H. Govil, "A deep residual convolutional neural network for mineral classification," *Adv. Space Res.*, vol. 71, no. 8, pp. 3186–3202, Apr. 2023.

[2] M. Shimoni, R. Haelterman, and C. Perneel, "Hyperspectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

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[3] H. Zhang, Y. Li, Y. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery using a dual-channel convolutional neural network," *Remote Sens. Lett.*, vol. 8, no. 5, pp. 438–447, May 2017.

[4] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[5] J. Zhu, L. Fang, and P. Ghamisi, "Deformable convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1254–1258, Aug. 2018.

[6] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[7] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[8] M. Ahmad et al., "Multi-head spatial–spectral mamba for hyperspectral image classification," *Remote Sens. Lett.*, vol. 16, no. 4, pp. 339–353, Apr. 2025.

[9] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8989–9003, Nov. 2023.

[10] H. Liu, W. Li, X.-G. Xia, M. Zhang, and R. Tao, "Multiarea target attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510613.

[11] H. Liu, W. Li, X.-G. Xia, M. Zhang, Z. Guo, and L. Song, "SegHSI: Semantic segmentation of hyperspectral images with limited labeled pixels," *IEEE Trans. Image Process.*, vol. 33, pp. 6469–6482, 2024.

[12] Y. Zhang, W. Li, W. Jia, M. Zhang, R. Tao, and S. Liang, "Cross-domain hyperspectral image classification based on bi-directional domain adaptation," 2025, *arXiv:2507.02268*.

[13] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[14] Y. Zhang, W. Li, M. Zhang, J. Han, R. Tao, and S. Liang, "SpectralX: Parameter-efficient domain generalization for spectral remote sensing foundation models," 2025, *arXiv:2508.01731*.

[15] C. Liu et al., "Boosting adversarial transferability for hyperspectral image classification using 3D structure-invariant transformation and weighted intermediate feature divergence," 2025, *arXiv:2506.10459*.

[16] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.

[17] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[18] J.-C. Burnel, K. Fatras, and N. Courty, "Generating natural adversarial hyperspectral examples with a modified Wasserstein GAN," 2020, *arXiv:2001.09993*.

[19] C. Shi, M. Zhang, Z. Lv, Q. Miao, and C.-M. Pun, "Universal object-level adversarial attack in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[20] J. Su, D. V. Vargas, and K. Sakurai, "One pixel attack for fooling deep neural networks," *IEEE Trans. Evol. Comput.*, vol. 23, no. 5, pp. 828–841, Oct. 2019.

[21] X. Peng, D. Zhou, G. Sun, and Y. Hu, "Adversarial system of one-pixel attack for hyperspectral image classification," *Research Square*, Aug. 2023. [Online]. Available: https://doi.org/10.21203/rs.3.rs-3221027/v1

[22] T. Bai, H. Wang, and B. Wen, "Targeted universal adversarial examples for remote sensing," *Remote Sens.*, vol. 14, no. 22, p. 5833, Nov. 2022.

[23] C. Shi, Y. Dang, L. Fang, Z. Lv, and M. Zhao, "Hyperspectral image classification with adversarial attack," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] C. Zhang et al., "An efficient general black-box adversarial attack approach based on multi-objective optimization for high dimensional images," *Comput. Electr. Eng.*, vol. 95, Oct. 2021, Art. no. 107402.

[25] W. Zhang, C. Jiang, J. Gui, and L. Dong, "Towards adversarial training under hyperspectral images," 2025, *arXiv:2510.01014*.

[26] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[27] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[28] R. L. Bassett, A. Van Dellen, and A. P. Austin, "Adversarial perturbations of physical signals," *Comput. Optim. Appl.*, vol. 90, no. 2, pp. 395–415, Mar. 2025.

[29] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[30] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[31] H. Li et al., "Sparse unmixing guided adversarial attack for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Aug. 25, 2025, doi: 10.1109/TCSVT.2025.3602604.

[32] X. Zhang, C. Sun, H. Han, and H. Sun, "Object-fabrication targeted attack for object detection," *Neurocomputing*, vol. 627, Apr. 2025, Art. no. 129561.

[33] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

[34] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Artificial Intelligence Safety and Security*. London, U.K.: Chapman & Hall, 2018, pp. 99–112.

[35] Z. Han, D. Hong, L. Gao, J. Yao, B. Zhang, and J. Chanussot, "Multimodal hyperspectral unmixing: Insights from attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[36] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[38] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1–11.

[39] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[40] K. Sun, F. Dong, W. Liu, Q. Wu, X. Sun, and W. Wang, "Hyperspectral image classification with spatial–spectral–channel 3-D attention and channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[41] X. Fei, S. Wu, J. Miao, G. Wang, and L. Sun, "Lightweight-VGG: A fast deep learning architecture based on dimensionality reduction and nonlinear enhancement for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 2, p. 259, Jan. 2024.

[42] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[43] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[44] H. Fang, J. Kong, B. Chen, T. Dai, H. Wu, and S.-T. Xia, "CLIP-guided generative networks for transferable targeted adversarial attacks," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 1–19.

[45] J. Zhang et al., "Anyattack: Towards large-scale self-supervised adversarial attacks on vision-language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 19900–19909.

[46] Z. Zhao, Z. Liu, and M. Larson, "On success and simplicity: A second look at transferable targeted attacks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 6115–6128.

[47] H. Wu, G. Ou, W. Wu, and Z. Zheng, "Improving transferable targeted adversarial attacks with model self-enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 24615–24624.

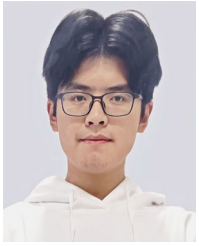
[48] B. Chen, J. Yin, S. Chen, B. Chen, and X. Liu, "An adaptive model ensemble adversarial attack for boosting adversarial transferability," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4466–4475.

[49] B. Tang, Z. Wang, Y. Bin, Q. Dou, Y. Yang, and H. T. Shen, "Ensemble diversity facilitates adversarial transferability," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 24377–24386.

[50] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.

[51] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9185–9193.

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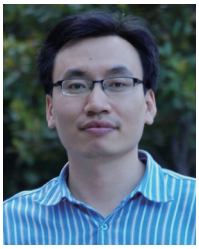
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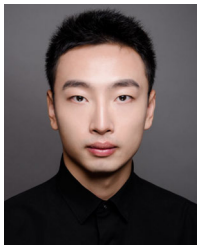
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